WASP-12b

R-0556

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-12_171224` · created 2026-07-23T14:27:11+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458111.75291 +/- 0.00045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.118 +/- 0.0085
Transit depth (Rp/Rs)^2	1.39 +/- 0.2 [%]
Orbital Inclination (inc)	79.81 +/- 2.48
Airmass coefficient 1 (a1)	1.0238 +/- 0.0088
Airmass coefficient 2 (a2)	-0.021 +/- 0.0056
Scatter in the residuals of the lightcurve fit is	0.86 %
Best Comparison Star	#3 - [419, 298]
Optimal Aperture	3.79
Optimal Annulus	10.88
Transit Duration (day)	0.1151 +/- 0.0084

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.118 ± 0.0085 vs 0.1178 (deviation 0.02σ)
Duration fit vs published	0.1151 d vs 0.125 d (deviation 1.05σ)
Mid-time O–C	-0.9 min
Depth SNR	12.4
Residual scatter	0.86 %

Light curve

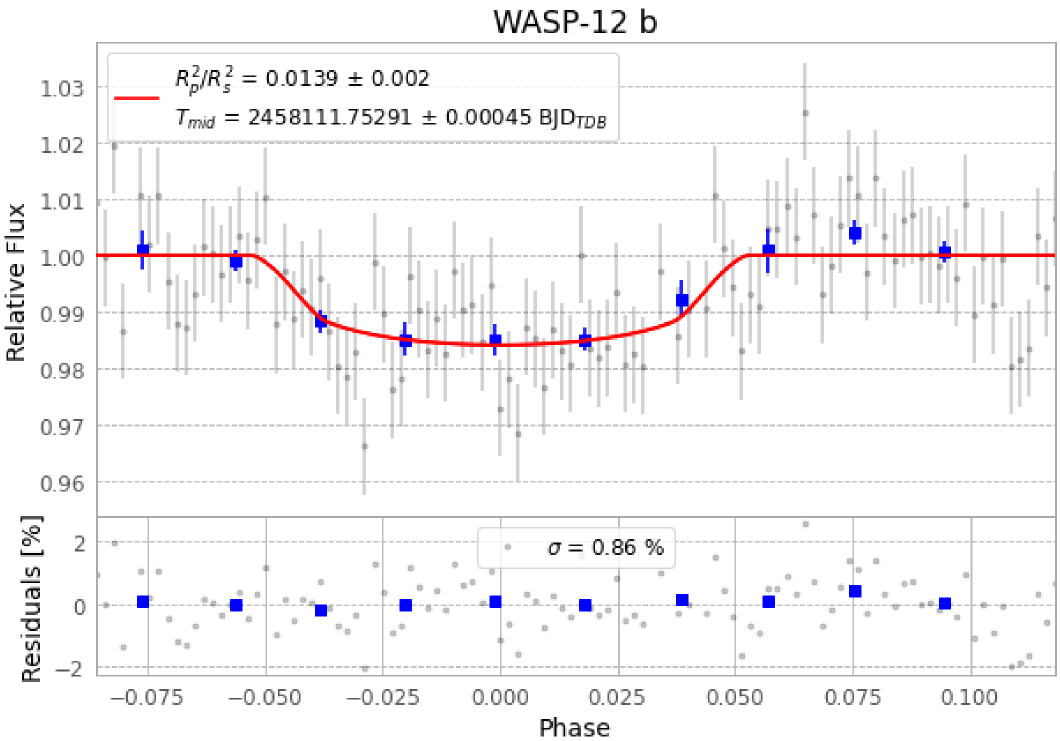

Figure 1 — FinalLightCurve_WASP-12 b_2017-12-23.png (SHA-256 4622ebff74659eb5…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [357, 219], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 6502086168259286…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494`

Data provenance

108 FITS frames (71 MB), `WASP-12171224033912.FITS` → `WASP-12171224090012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-171224.log (5a9d55640050…), FinalLightCurve_WASP-12 b_2017-12-23.png (4622ebff7465…), FinalLightCurve_WASP-12 b_2017-12-23.csv (71fe7acb4ca1…)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)